PID Control: Overview and pole placement

R0005E - Measurement and Feedback Control

Wolfgang Birk

Department of Computer Science, Electrical and Space Engineering

LULEÅ UNIVERSITY OF TECHNOLOGY

PID Control

Due to the advantages of closed loop systems, we will now only focus on such systems and namely on feedback control systems.

Until now, the choice of the closed loop controller D_{cl} has been a gain K. This means that the control action is proportional (**P**) to the control error.

We will now extend this notion, by adding

- **I:** integral action (Considering the history of the control error).
- **D:** derivative action (Considering the direction of change of the control error).
- **ID:** a combination of both integral and derivative action (ID)

This yields four controller architectures: P, PI, PD and PID. The PID can be given in the following form:

$$D(s) = \frac{U(s)}{E(s)} = k_P + \frac{k_I}{s} + k_D s$$

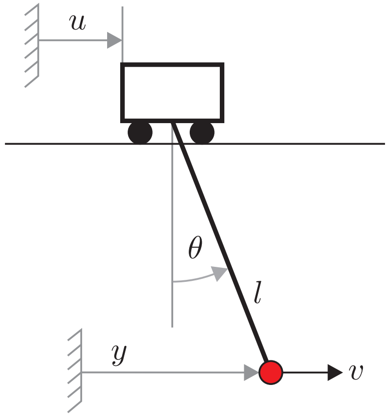
By setting some parameters to zero, it is possible to reduce the PID to the other three controller architectures.

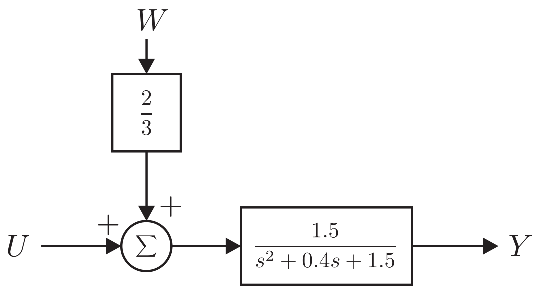
Note: This description is not the only possible for a PID controller. In other textbooks, different implementations are used, like for example

$$D(s) = k_P \left(1 + \frac{1}{T_I s} + T_D s\right)$$

Example system: traverse crane

In order to explore these controllers the following traverse crane example will be used:





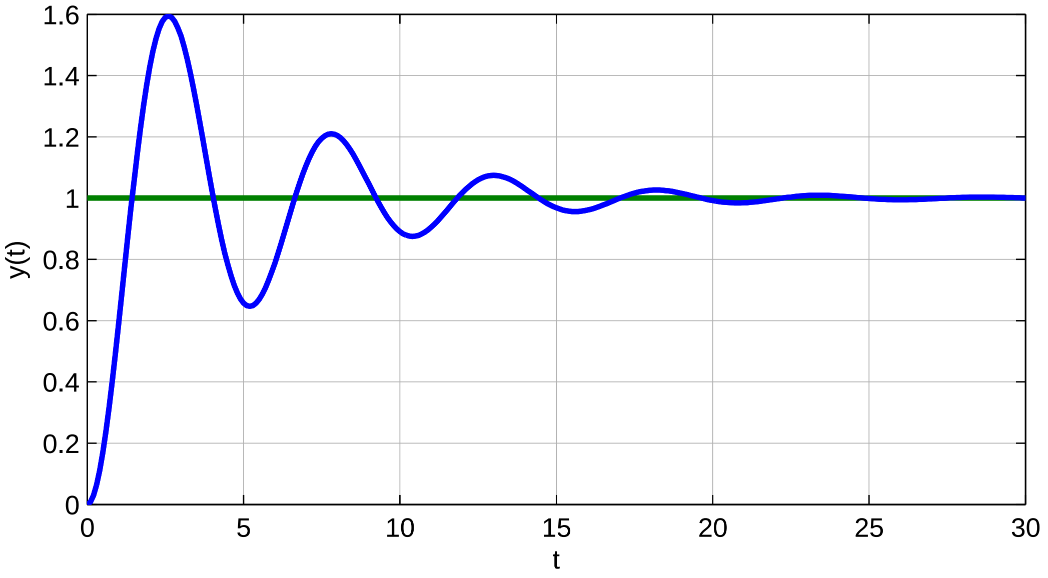
For this crane we will analyze how the controller is performing with respect to:

- **Regulation:** Here we will only focus on the disturbance rejection ability and not the ability to attenuate sensor noise.
  $\Rightarrow$ Analysis of $T_W(s) = \frac{Y(s)}{W(s)}$ or $S_W(s) = \frac{E(s)}{W(s)}$
- **Tracking:** Here we will analyse how well the system is able to follow a set point change, given in terms of a step input.
  $\Rightarrow$ Analysis of $T(s) = \frac{Y(s)}{R(s)}$ or $S(s) = \frac{E(s)}{R(s)}$

Note: We will have a look at the DC-gain of these transfer functions. It is desired to achieve $T(0) \to 1$ and $T_W(0) \to 0$.

Model analysis: traverse crane

The crane is a 2^{nd} order system which is lightly damped. This can be seen from the step response of the system:



The parameters of the system derived from the transfer function are:

$$\omega_n = \sqrt{1.5}\frac{rad}{s} = 1.22\frac{rad}{s}$$

$$\zeta = 0.164$$

$$t_r = \frac{1.8}{\omega_n} = 1.47s$$

$$t_p = 2.81s$$

$$t_s = \frac{4.6}{\zeta\omega_n} = 23s$$

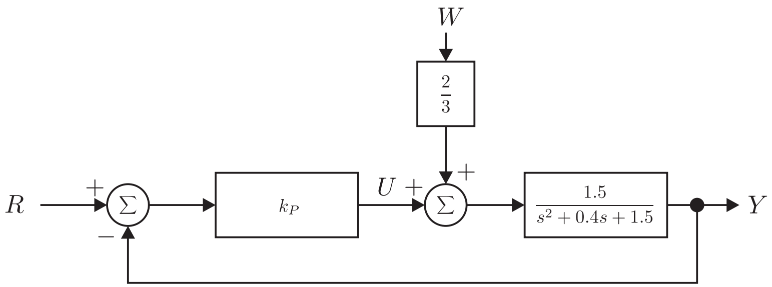
$$M_P = e^{-\zeta\frac{\pi}{\sqrt{1-\zeta^2}}} = 0.57$$

P Control

Controller:

$$D(s) = k_P$$

Closed loop system:



Closed loop transfer functions:

$$T(s) = \frac{G(s)D(s)}{1+G(s)D(s)} = \frac{1.5k_P}{s^2+0.4s+1.5+1.5k_P}$$

$$T_W(s) = \frac{G_W(s)G(s)}{1+G(s)D(s)} = \frac{1}{s^2+0.4s+1.5+1.5k_P}$$

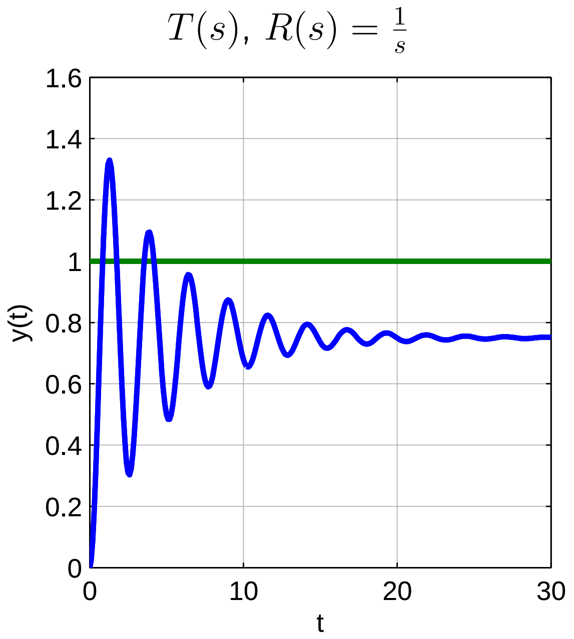
DC-Gains:

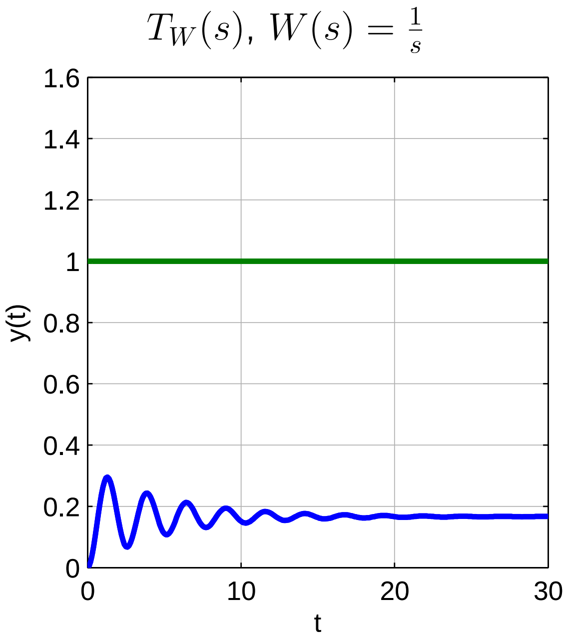
We evaluate the transfer functions $T(s)$ and $T_W(s)$ for $s = 0$:

$$T(0) = \frac{1.5k_P}{1.5+1.5k_P}, \qquad T_W(0) = \frac{1}{1.5+1.5k_P}$$

Simulation of the closed loop system for unit step inputs:

We tune the P-Controller to have the gain $k_P = 3$.





Analysis:

- **DC Gains:** $T(0) = \frac{4.5}{6}$ and $T_W(0) = \frac{1}{6}$.

- **Steady state control error:**

$$
\begin{aligned}
e_{ss} &= \lim_{s \to 0} sE(s) \\
&= \lim_{s \to 0} s \left\{ R(s) - T(s)R(s) - T_W(s)W(s) \right\} \\
&= \lim_{s \to 0} s \left\{ (1 - T(s))R(s) - T_W(s)W(s) \right\} \\
&= \frac{1.5}{1.5 + 1.5k_P} R(0) - \frac{1}{1.5 + 1.5k_P} W(0)
\end{aligned}
$$

There, $R(0)$ and $W(0)$ are the steady state values for $r(t)$ and $w(t)$

Closed loop characteristic equation:

Remember that the roots of the characteristic equation determine the poles of the closed loop system, which in turn govern the dynamic behavior.

$$
\begin{aligned}
& 1 + G(s)D(s) = 0 \\
\Rightarrow \qquad & s^2 + 0.4s + 1.5 + 1.5k_P = 0
\end{aligned}
$$

Thus, the gain $k_P > 0$ of the controller affects the pole locations and are given by

$$
s_{1,2} = -0.2 \pm \sqrt{\underbrace{0.04 - (1.5 + 1.5k_P)}_{<0}}
$$

Conclusion: The controller can affect ω_n and ζ but not σ and yields a constant tracking error and regulation error.

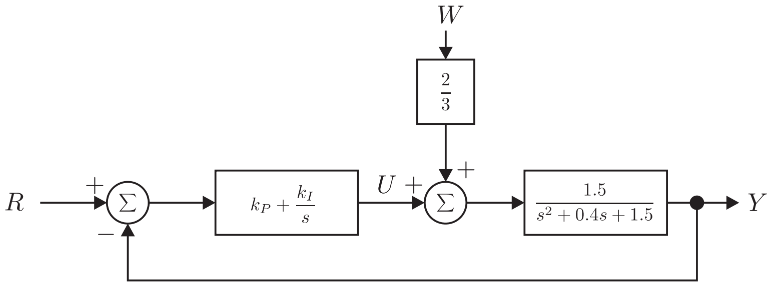
PI Control

Controller:

$$D(s) = k_P + \frac{k_I}{s}$$

Here we have two parameters to adjust. Moreover, the controller has the ability to consider the history of the control error. Thus, this could help to remove a constant control error!

Closed loop system:



Closed loop transfer functions:

$$T(s) = \frac{G(s)D(s)}{1+G(s)D(s)} = \frac{1.5k_P s + 1.5k_I}{s^3 + 0.4s^2 + (1.5+1.5k_P)s + 1.5k_I}$$

$$T_W(s) = \frac{G_W(s)G(s)}{1+G(s)D(s)} = \frac{s}{s^3 + 0.4s^2 + (1.5+1.5k_P)s + 1.5k_I}$$

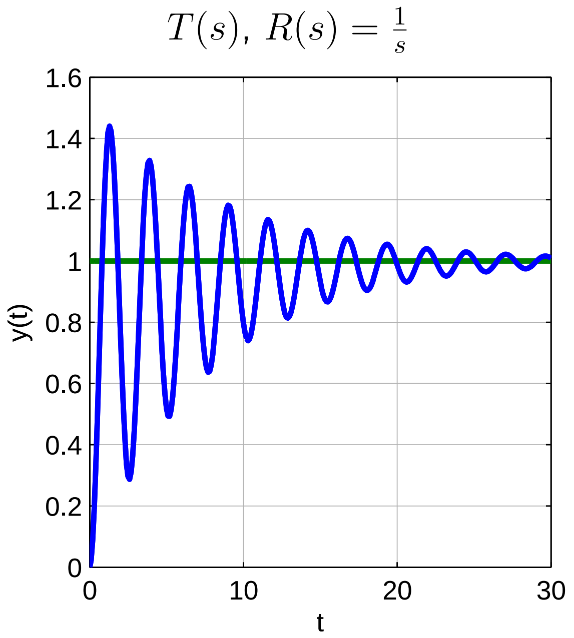
DC-Gains:

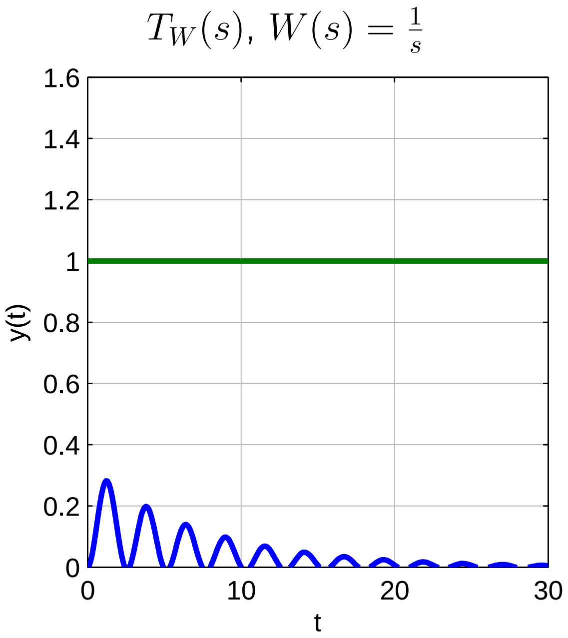
We evaluate the transfer functions $T(s)$ and $T_W(s)$ for $s = 0$:

$$T(0) = \frac{1.5k_I}{1.5k_I} = 1, \qquad T_W(0) = 0$$

Simulation of the closed loop system for unit step inputs:

We tune the PI-Controller to have the gains $k_P = 3$ and $k_I = 0.6$.





PI Control (3)

Analysis:

- **DC Gains:** $T(0) = 1$ and $T_W(0) = 0$.
- **Steady state control error:**

$$\begin{aligned} e_{ss} &= \lim_{s\to 0} sE(s) \\ &= \lim_{s\to 0} s\left\{R(s) - T(s)R(s) - T_W(s)W(s)\right\} \\ &= \lim_{s\to 0} s\left\{(1 - T(s))R(s) - T_W(s)W(s)\right\} \\ &= 0 \cdot R(0) - 0 \cdot W(0) \end{aligned}$$

Closed loop characteristic equation:

Remember that the roots of the characteristic equation determine the poles of the closed loop system, which in turn govern the dynamic behavior.

$$\begin{aligned} & 1 + G(s)D(s) = 0 \\ \Rightarrow \qquad & s^3 + 0.4s^2 + (1.5 + 1.5k_P)s + 1.5k_I = 0 \end{aligned}$$

Thus, the gains $k_P > 0$ and $k_I > 0$ of the controller affect the pole locations, but not all parameters of the characteristic equation. Additionally, there are now three poles.

Conclusion: The controller yields a zero tracking error and regulation error. But it introduces oscillations.

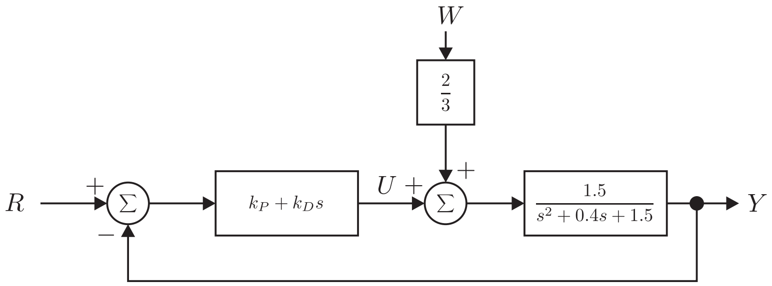
PD Control

Controller:

$$D(s) = k_P + k_D s$$

Here we have again two parameters to adjust. Moreover, the controller has the ability to consider the direction of change of the control error. Thus, the controller could anticipate what is going to happen!

Closed loop system:



Closed loop transfer functions:

$$T(s) = \frac{G(s)D(s)}{1 + G(s)D(s)} = \frac{1.5k_P + 1.5k_D s}{s^2 + (0.4 + 1.5k_D)s + 1.5 + 1.5k_P}$$

$$T_W(s) = \frac{G_W(s)G(s)}{1 + G(s)D(s)} = \frac{1}{s^2 + (0.4 + 1.5k_D)s + 1.5 + 1.5k_P}$$

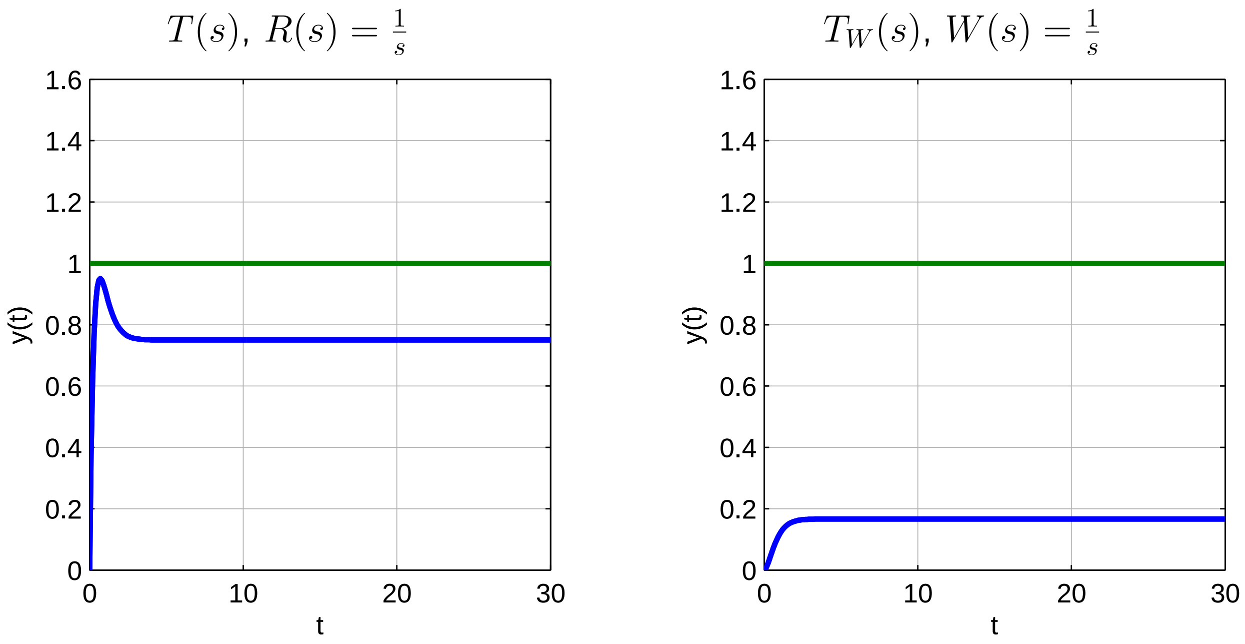
DC-Gains:

We evaluate the transfer functions $T(s)$ and $T_W(s)$ for $s = 0$:

$$T(0) = \frac{1.5k_P}{1.5 + 1.5k_P}, \qquad T_W(0) = \frac{1}{1.5 + 1.5k_P}$$

Simulation of the closed loop system for unit step inputs:

We tune the PD-Controller to have the gains $k_P = 3$ and $k_D = 3$.



Analysis:

- **DC Gains:** $T(0) = \frac{4.5}{6}$ and $T_W(0) = \frac{1}{6}$.
- **Steady state control error:**

$$\begin{aligned} e_{ss} &= \lim_{s \to 0} sE(s) \\ &= \lim_{s \to 0} s \left\{ R(s) - T(s)R(s) - T_W(s)W(s) \right\} \\ &= \lim_{s \to 0} s \left\{ (1 - T(s))R(s) - T_W(s)W(s) \right\} \\ &= \frac{1.5}{1.5 + 1.5k_P} R(0) - \frac{1}{1.5 + 1.5k_P} W(0) \end{aligned}$$

Closed loop characteristic equation:

Remember that the roots of the characteristic equation determine the poles of the closed loop system, which in turn govern the dynamic behavior.

$$\begin{aligned} & & 1 + G(s)D(s) &= 0 \\ \Rightarrow & & s^2 + (0.4 + 1.5k_D)s + (1.5 + 1.5k_P) &= 0 \end{aligned}$$

Thus, the gains $k_P > 0$ and $k_D > 0$ of the controller affect the pole locations, and all parameters of the characteristic equation depend on the controller parameters.

Conclusion: The controller can affect both ω_n and ζ of the system but yields a constant non-zero tracking error and regulation error.

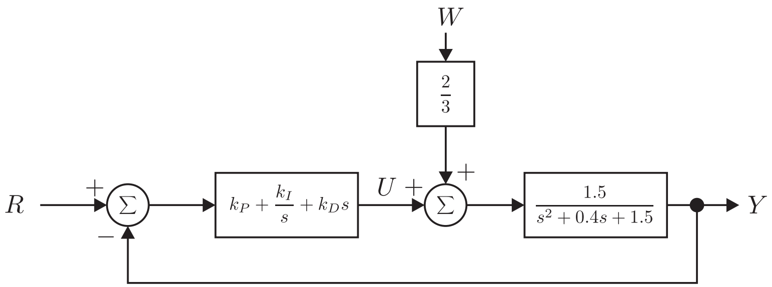
PID Control

Controller:

$$D(s) = k_P + \frac{k_I}{s} + k_D s$$

Now we combine the properties of the PD with the ones of the PI controller. This should both address the constant controller and the still give us access to the damping coefficient of the closed loop system.

Closed loop system:



Closed loop transfer functions:

$$T(s) = \frac{G(s)D(s)}{1+G(s)D(s)} = \frac{1.5k_D s^2 + 1.5k_P s + 1.5k_I}{s^3 + (0.4+1.5k_D)s^2 + (1.5+1.5k_P)s + 1.5k_I}$$

$$T_W(s) = \frac{G_W(s)G(s)}{1+G(s)D(s)} = \frac{s}{s^3 + (0.4+1.5k_D)s^2 + (1.5+1.5k_P)s + 1.5k_I}$$

PID Control (2)

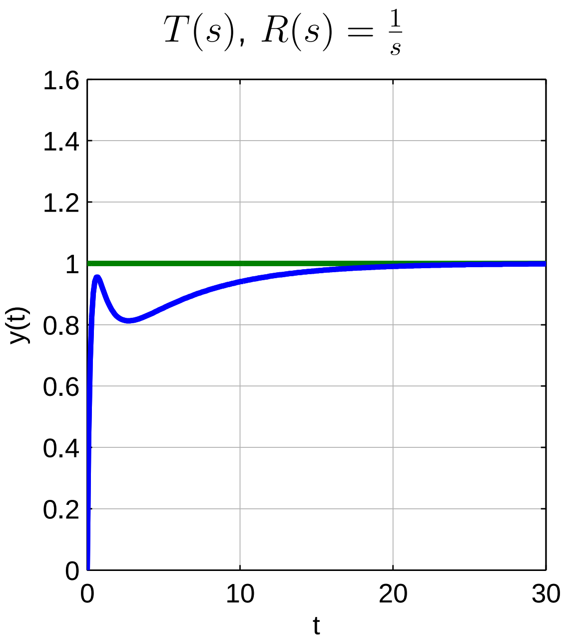
DC-Gains:

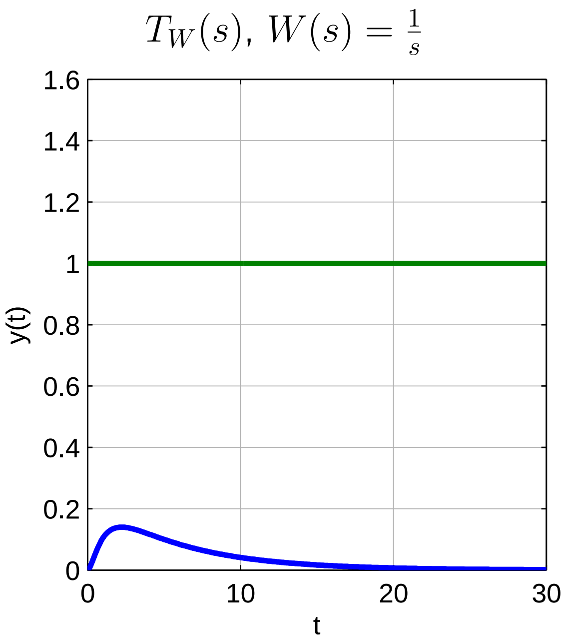
We evaluate the transfer functions $T(s)$ and $T_W(s)$ for $s = 0$:

$$T(0) = \frac{1.5k_I}{1.5k_I} = 1, \qquad T_W(0) = 0$$

Simulation of the closed loop system for unit step inputs:

We tune the PID-Controller to have the gains $k_P = 3$, $k_I = 0.6$ and $k_D = 3.6$.





Analysis:

- **DC Gains:** $T(0) = 1$ and $T_W(0) = 0$.
- **Steady state control error:**

$$\begin{aligned} e_{ss} &= \lim_{s\to 0} sE(s) \\ &= \lim_{s\to 0} s\left\{R(s) - T(s)R(s) - T_W(s)W(s)\right\} \\ &= \lim_{s\to 0} s\left\{(1 - T(s))R(s) - T_W(s)W(s)\right\} \\ &= 0 \cdot R(0) - 0 \cdot W(0) \end{aligned}$$

Closed loop characteristic equation:

Remember that the roots of the characteristic equation determine the poles of the closed loop system, which in turn govern the dynamic behavior.

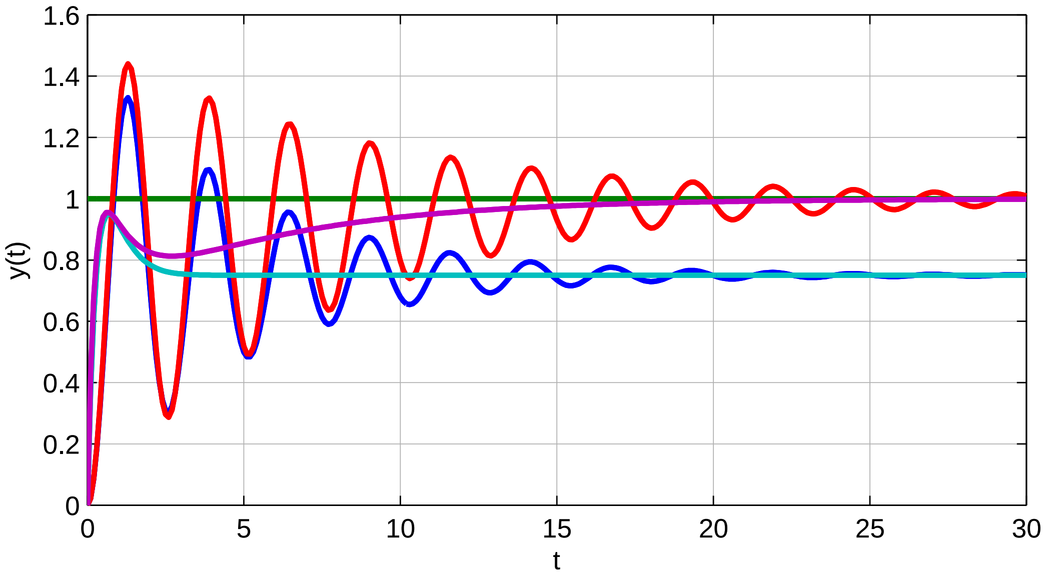
$$\begin{aligned} & 1 + G(s)D(s) = 0 \\ \Rightarrow \qquad & s^3 + (0.4 + 1.5k_D)s^2 + (1.5 + 1.5k_P)s + 1.5k_I = 0 \end{aligned}$$

Thus, the gains $k_P > 0$, $k_I > 0$ and $k_D > 0$ of the controller affect the pole locations, and all parameters of the characteristic equation depend on the controller parameters, even though there are three poles now.

Conclusion: The controller can affect both ω_n and ζ of the system and yields a zero tracking error and regulation error. Perfect!

Combined simulation results

Just to summarize the controllers with their response $T(s)$ and their parameters we get the following plot:



- P-Controller (Blue): $k_P = 3$
- PI-Controller (Red): $k_P = 3$, $k_I = 0.6$
- PD-Controller (Cyan): $k_P = 3$, $k_D = 3$
- PID-Controller (Magenta): $k_P = 3$, $k_I = 0.6$, $k_D = 3.6$

Pole placement

We have seen that the controller parameters have an affect on the location of the poles. Depending on the complexity of the controller (the larger the number of parameters in the controller, the larger is its complexity), the locations can be uniquely defined.

Pole placement:

1. Derive the desired characteristic polynomial $\hat{P}(s)$
2. Derive the resulting characteristic polynomial for the closed loop system with the controller. $P(s)$
3. Compare the coefficients of the desired and the resulting characteristic polynomial, and derive equation for the controller parameters so that $P(s) = \hat{P}(s)$.
4. Solve the algebraic equations for the controller parameters.

Important notes:

- The desired characteristic polynomial is derived from the specifications of the closed loop system.
- The controller needs to have sufficiently high complexity, so that the poles can be placed freely.

Additional material

The following two videos give a practical introduction to PID controllers and what they do.

- PID Control - A brief introduction
  `https://www.youtube.com/watch?v=UR0hOmjaHp0`
- Simple Examples of PID Control
  `https://www.youtube.com/watch?v=XfAt6hNV8XM`